

Chapter 4
MOVING CHARGES AND MAGNETISM
SECTION A

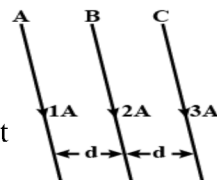
MCQ

1. An electron having energy 10eV is circulating in path having magnetic field 1.0×10^{-4} Tesla, the speed of the electron will be
(a) $1.9 \times 10^6 m/s$ (b) $3.8 \times 10^6 m/s$ (c) $1.9 \times 10^{12} m/s$ (d) $3.8 \times 10^{12} m/s$
2. A proton passing from a fixed place with constant velocity. If E and B are electric and magnetic field respectively, then which statement is true?
(a) $E \neq 0, B = 0$ (b) $B \neq 0, E = 0$ (c) $E \neq 0, B \neq 0$ (d) None of the above
3. A solenoid of 1.5 metre length and 4.0 cm diameter possesses 10 turns per cm. A current 5 amp is flowing through it. The magnetic field the axis inside the solenoid is
(a) $2\pi \times 10^{-3} T$ (b) $2\pi \times 10^6 T$ (c) $4\pi \times 10^{-3} T$ (d) $2\pi \times 10^3 T$
4. A particle having charge 100 times that of a electron is revolving in a circular path of radius 0.8 m with one rotation per second. The magnetic field produced at the centre is
(a) $10^{-7} \mu_0$ (b) $10^{-17} \mu_0$ (c) $10^{-6} \mu_0$ (d) $10^{-15} \mu_0$
5. Biot-Savart law indicates that the moving electrons (velocity v) produce a magnetic field B such that
(a) $B \perp v$.
(b) $B \parallel v$.
(c) it obeys inverse cube law.
(d) it is along the line joining the electron and point of observation.
6. The coil of a moving coil galvanometer is wound over a metal frame in order to
(a) Reduce hysteresis
(b) Increase the sensitivity
(c) Increase moment of inertia
(d) Provide electromagnetic damping

7. Two wires of same length are shaped into a square of side 'a' and circle of radius 'r'. If they carry same current, the ratio of the magnetic moment is

(a) $2: \pi$ (b) $\pi: 2$ (c) $\pi: 4$ (d) $4: \pi$

8. Three infinitely long parallel straight current carrying wires A, B and C are kept equal distance from each other as shown in the figure given. The wire C experiences net force **F**. The net force on wire C, when the current in wire A is reversed will be



(a) Zero (b) $\frac{F}{2}$ (c) F (d) 2F

9. The nature of parallel and anti-parallel currents are
a. Parallel current repels and anti-parallel current attract
b. Parallel current attracts and anti-parallel current repel
c. Both current attract
d. Both current repel.

10. Intensity of magnetic field due to the bar magnet at a point inside a hollow steel-box is
a. Less than outside b. Same as outside c. More than outside d. Zero

1 MARK QUESTIONS

11. Write the expression, in a vector form, for the Lorentz magnetic force $\vec{F}$ due to a charge moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$. What is the direction of the magnetic force?
12. What is figure of merit of a galvanometer?
13. In a certain arrangement, a proton does not get deflected while passing through a magnetic field region. State the condition under which it is possible.
14. Using the concept of force between two infinitely long parallel current carrying conductors, define one ampere of current. (All India 2014)
15. Write the expression, in a vector form, for the Lorentz magnetic force $\vec{F}$ due to a charge moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$. What is the direction of the magnetic force? (CBSE 2014)
16. Why should be the spring/suspension wire in a moving coil galvanometer have low torsional constant? (All India 2008)
17. What is the underlying principle of a moving coil galvanometer? (CBSE 2019)

18. A coil of area A carrying a steady current I has a magnetic moment $\vec{m}$ associated with it. Write the relation between $\vec{m}$, I and A in vector form. (Comptt 2016)

2-MARK QUESTIONS

19. A current carrying loop is free to turn in a uniform magnetic field B . Under what conditions, will the torque acting on it be (i) minimum and (ii) maximum?
20. A current carrying loop is free to turn in a uniform magnetic field B . Under what conditions, will the torque acting on it be (i) minimum and (ii) maximum?
21. A steady current (I_1) flows through a long straight wire. Another wire carrying steady current (I_2) in the same direction is kept close and parallel to the first wire. Show with the help of a diagram how the magnetic field due to current (I_1) exerts a magnetic force on the second wire. Write expression for this force. (All India 2013)
22. A moving coil galvanometer, whose coil resistance is $100\ \Omega$, shows full scale deflection when 1mV is put across it. How can it be converted into a voltmeter of range $(0-1\text{V})$?
23. How do convert a galvanometer into an ammeter? Why is an ammeter always connected in series? (AI 2005)
24. What is the behaviour of magnetic field lines due to diamagnetic material?

3-MARKS QUESTIONS

25. A long straight wire in the horizontal plane carries a current of 50 A in north to south direction. Give the magnitude and direction of B at a point 2.5 m east of the wire.
26. Write the expression for Lorentz magnetic force on the particle of charge q moving with velocity v in a magnetic field B . Show that no work is done by this force on the charged particle.
27. Derive the expression for the torque τ acting on a rectangular current loop of area A placed in a uniform magnetic field B . Show that $\vec{\tau} = \vec{m} \times \vec{B}$ where $\vec{m}$ is the magnetic moment of the current loop given by $\vec{m} = I\vec{A}$.
28. Describe the working principle of a moving coil galvanometer. Why is it necessary to use (i) a radial magnetic field and (ii) a cylindrical soft iron core in a galvanometer? Write the expression for current sensitivity of the galvanometer. Can a galvanometer as such be used for measuring the current? Explain.

5-MARKS QUESTIONS

29. (i) Explain using a labelled diagram the principle and working of a moving coil galvanometer.
What is the function of (a) uniform radial magnetic field (b) soft iron core?
(ii) Define the terms (a) Current sensitivity (b) Voltage sensitivity
(iii) Explain why does increasing the current sensitivity not necessarily increase the voltage sensitivity

Assertion-Reason

Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false and R is also false.
30. **Assertion** Difference between an electric line and magnetic line of force is that electric lines of force are discontinuous and the magnetic field lines are continuous.
Reason Electric lines of forces do not exist inside a charged conductor but magnetic lines exist inside a magnet.
31. **Assertion** A current carrying solenoid behaves like a bar magnet.
Reason The circular loop in which the direction of current is clockwise behaves like the South Pole and the one having anticlockwise current behaves like the North Pole.
32. **Assertion** Permanent magnets retain their ferromagnetic property for a long period of time.
Reason Steel is a diamagnetic material.
33. **Assertion** When a bar magnet is hung freely it points toward geographical poles.
Reason Magnetic field lines do not intersect.
34. **Assertion** A diamagnetic specimen would move towards the weaker region of the field.
Reason A diamagnetic specimen is repelled by a magnet.
35. **Assertion** Motion of electron around a positively charged nucleus is different from the motion of a planet around the sun.
Reason The force acting in both the cases is same in nature.

36.**Assertion** Two parallel conducting wires carrying currents in same direction, come close to each other.

Reason Parallel currents attract and anti-parallel currents repel.

37.**Assertion** A galvanometer cannot as such be used as an ammeter to measure the current across a given section of the circuit.

Reason For this it must be connected in series with the circuit.

38.**Assertion** Magnetic lines of force form continuous closed loops whereas electric lines of force do not.

Reason Magnetic poles always occur in pairs as north pole and south pole.

39.**Assertion** An electron moving along the direction of magnetic field experiences no force.

Reason The force on electron moving along the direction of magnetic field is $F = qVB \sin 0^\circ = qVB$

40. **Assertion** Iron is not a magnet.

Reason Iron is diamagnetic substance

Case study based question

41. Force on a moving charge

Stationary charge creates an electric field but a moving charge creates a magnetic field also that can affect other moving charges. It was observed by Oersted. He saw that if a magnetic needle is placed near current carrying wire it shows slight deflection. The direction of the magnetic field can be determined by using the right hand thumb rule. Magnetic field is the space around a magnet, a current carrying conductor up to which it can attract or repel magnetic material. Force on a moving charge in magnetic field is given by the formula

$$\vec{F} = q (\vec{v} \times \vec{B})$$

Where q = charge on particle

v = velocity of charge particle

B = magnetic field and

F = magnetic force

When a current carrying conductor is placed in an external magnetic field, it experiences a mechanical force. A conductor of length l carrying current I held in a magnetic field B at an angle θ with it, experiences a force given by $F = I l B \sin \theta$.

1. Moving charge can create
(a) Electric field (b) magnetic field
(c) Both electric and magnetic field (d) none of them
2. If a current is flowing from south to north in a straight wire what will be the direction of magnetic field to its left side
(a) Outward (b) inward
(c) Towards right (d) towards left
3. Which of the following cannot be the source of magnetic field?
(a) Current carrying wire (b) moving electron
(c) Moving proton (d) stationary charge
4. Force acting on a conductor of length 10 m carrying a current of 6A kept perpendicular to the magnetic field of 2T is
(a) 60N (b) 120N (c) 90N (d) 100N
5. A straight wire is of mass 100 g and length 1.0 m carrying a current of 5A. It is suspended in a uniform horizontal magnetic field **B**. The magnitude of **B** is (assume that $g = 9.8\text{ms}^{-2}$)
(a) 0.196T (b) 0.46T (c) 0.15T (d) 3.26T

ANSWER KEY

SECTION A

MCQ TYPE (1MARK EACH)

- 1.(a) $1.9 \times 10^6 \text{ m/s}$
- 2.(c) $E \neq 0, B \neq 0$
3. (a) $2\pi \times 10^{-3} \text{ T}$
4. (b) $10^{-17} \mu_0$
5. (a) $B \perp v$
6. (d) Provide electromagnetic damping.
7. (c) $\pi: 4$
8. (a) Zero
9. (b) Parallel current attracts and anti-parallel current repel
10. d) Intensity of magnetic field inside a steel box is zero. No magnetic lines of force pass through the box.

1 MARK

11. $\vec{F} = q(\vec{v} \times \vec{B})$. The direction of force is perpendicular to both v and B .
12. It is defined as the current which produces a deflection of one scale division in the galvanometer
13. v is parallel or antiparallel to B
14. Using $F/l = \mu_0 I_1 I_2 / 2\pi r$. One Ampere is the current which when flowing through two infinitely long, straight parallel conductors of negligible cross section held 1m apart in vacuum develop force of $2 \times 10^{-7} \text{ N/m}$ of the conductors
15. $F = q(\vec{v} \times \vec{B})$, The force acts in a direction given by the thumb of the right hand with fingers circling from $\vec{v}$ to $\vec{B}$.
16. Low torsional constant is basically required to increase the current sensitivity in an MCG.
17. A current carrying coil, in the presence of magnetic field experiences a torque which produces proportionate deflection. Deflection $\propto$ Torque
18. Magnetic Moment $\vec{m} = I \vec{A}$

2-MARKS

19. Minimum potential energy = $-MB$ when $\theta = 0$ (most stable position)

Maximum potential energy = MB when $\theta = 180^\circ$ (most unstable position)

20. loop will come to equilibrium when torque is zero

$$\begin{aligned}\text{So, } \tau &= 0 = M \times B \\ &= MB \sin \theta \\ &= NiAB \sin \theta\end{aligned}$$

So, angle between area vector and field is zero than only θ is 0 so that $\sin \theta = 0$.
So, for area vector along field direction plane is perpendicular to the field.

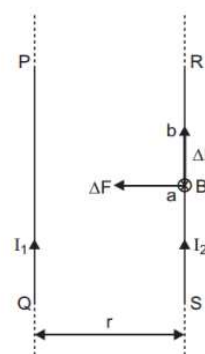
21.

Let the given wires PQ and RS separated by a distance 'r'. As wires carry current, they produce magnetic field around them. wire Produces magnetic field of intensity B at P on RS, which equal to $\mu_0 I_1 / 2\pi r$ The wire RS carrying current I_2 and placed Magnetic field of P Q Experiences a force given by

$$F = B I_2 l = (\mu_0 I_1 I_2 / 2\pi r). l$$

Similarly, also exert an equal and opposite force (attractive for like parallel currents) on the wire PQ, which is equal to

$$F = B I_1 l = (\mu_0 I_1 I_2 / 2\pi r). l \quad \text{N/m}$$



the
The
is
in a
RS

$$22. I_g = V/G = 1 \times 10^{-3} / 100 = 10^{-5} \text{ A}$$

$$\begin{aligned}V &= I_g G + I_g R, \\ &= 10^{-5} (100 + R)\end{aligned}$$

$$R = 100000 - 100 = 99900 \text{ ohms.}$$

23. We know that an ammeter has very low resistance. Thus, in order to convert a galvanometer into an ammeter, a very low shunt resistance must be connected in parallel to galvanometer resistance R_G so that the overall resistance of the circuit becomes very small. Thus, a galvanometer can be converted into an ammeter by adding a low shunt resistance in parallel to the galvanometer. It is connected in series so that whole of electric current, which it has to measure, passes through it.

24. A diamagnetic material tends to move from stronger to weaker regions of the magnetic field and hence, decrease the number of lines of magnetic field passing through it. Relative permeability is 1.

25.

3-MARKS

26. Current, $I=50\text{A}$, Distance, $r = 2.5 \text{ m}$, Magnetic field, $B = 2\pi r\mu_0 I$, $B = 4.0 \times 10^{-6} \text{ T}$

According to Maxwell's right hand, the direction of field is upward.

27. Lorentz magnetic force $F = q (V \times B)$. As the magnetic force F acts in the direction perpendicular to the direction of velocity V or the direction of motion of the charge particle, so the work done is zero. So, $W = F \cdot ds = F ds \cos \theta = f ds \cos 90 = 0$

28. Torque on a rectangular current loop in a uniform magnetic field

Let I = current through the coil

a, b = sides of the rectangular loop, $PQ = b$ & $QR = a$

$A = ab$ = area of the loop

n = number of turns in the loop

B = magnetic field

θ = angle between magnetic field

B and area vector A

Force exerted on the arm PQ inward

$$F_1 = I b B \quad [F = ILB]$$

Force exerted on the arm RS outward

$$F_3 = I b B \quad F_1 = F_3$$

Therefore, two equal and opposite forces form a couple which exerts a torque.

Therefore, Magnitude of the torque on the loop is,

$$\begin{aligned} \tau &= F_1 \times (a/2) \sin \theta + F_3 (a/2) \sin \theta \\ &= (F_1 + F_3) \cdot (a/2) \sin \theta \\ &= (IbB + IbB) \cdot (a/2) \sin \theta \end{aligned}$$

$$\Rightarrow \tau = 2IbB (a/2) \sin \theta$$

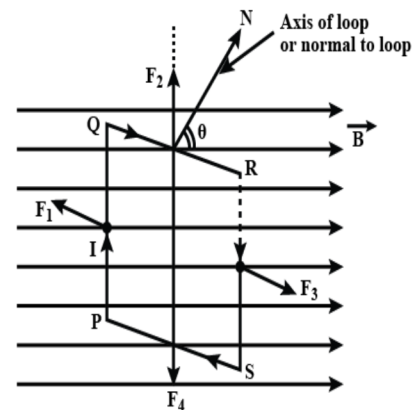
$$\Rightarrow = IabB \sin$$

$$= IAB \sin \theta$$

magnetic moment of the current loop is

$$M = IA$$

$$\tau = MB \sin \theta = \tau = M \times B$$



If loop has n turns then $M = nIA$

$$\tau = nIAB \sin\theta$$

when $\theta = 90^\circ$ then $\tau_{\max} = nIAB$

when $\theta = 0^\circ$ then $\tau = 0$.

29. (i) It keeps the magnetic field line normal to the area vector of the coil

(ii) The cylindrical soft iron core, when placed inside the coil of a galvanometer, makes the magnetic field stronger and radial in the space between it and pole pieces, such that whatever the position of the rotation of the coil maybe, the magnetic field is always parallel to its plane.

Current sensitivity is defined as the deflection produced by the galvanometer when unit current is passed through its coil.

$$I_s = \phi/I = NBA/k \text{ radian/ampere or division } A^{-1}.$$

No, the galvanometer cannot be used to measure current. it can only detect current but cannot measure as it is not calibrated. The galvanometer coil is likely to be damaged by currents in the (mA/A) range.

5-MARKS

30. (i) A galvanometer is an electromechanical instrument used for the detection of electric currents flowing through electrical circuits. It is a very sensitive instrument which cannot be used for the measurement of large currents. It works on the principle of conversion of electrical energy into mechanical energy while a current is flowing in a magnetic field as it experiences a magnetic torque and hence rotates through an angle proportional to the current flowing through it.

Let I = the current flowing through coil,

B = magnetic field parallel to the coil, and

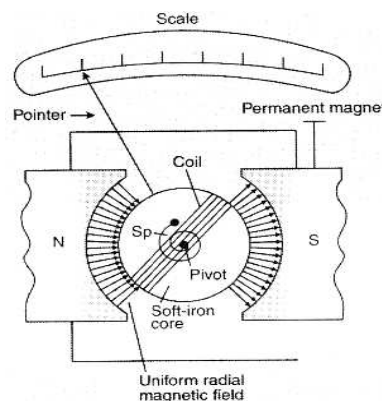
A = area of the coil.

Deflection acting on the coil is

$$\tau = NIBA \sin 90^\circ = NIBA \quad [\because \sin 90^\circ = 1]$$

(a) By making a uniform radial magnetic field through a coil, the magnetic field lines become perpendicular to the magnetic moment of a galvanometer.

$$\tau = k\Phi$$



$$NIBA = k\Phi$$

$$\Phi = (NBAI/k)$$

$$\therefore \Phi \propto I$$

(b) Soft Iron core can make the electromechanical field radial which in turn would increase the magnetic field.

(ii) Definition for the terms :

(a) Current Sensitivity- It is deflection produced per unit current flowing across the galvanometer

$$I_s = \theta/I = NBA/K$$

(b) Voltage Sensitivity- It is the minimum change in voltage which produces change in the output of the galvanometer.

$$V_s = \theta/V = \theta/IR = NBA/KR$$

(iii) Since Voltage sensitivity decreases with the increase in resistance of the coil and the effect of an increase in the number of turns is hence nullified in case of voltage sensitivity so there is no change of voltage sensitivity, whenever there is a change in current sensitivity.

Assertion-Reason

30.(a) Since no electric lines of forces exist inside a charged body, the electric lines of force only travel from positive to negative charge and are discontinuous. Secondly, magnetic lines of force travel from north to south pole and inside the magnet they are from south pole to north pole hence continuous.

31.(a) A solenoid is a type of electromagnet formed by a helical coil of wire whose length is substantially greater than its diameter, its two ends can be visualised as two coils.

32.(c) Permanent magnets retain their ferromagnetic property for a long period of time and steel is a paramagnetic material.

33.(b) It is the property of a magnet to rest in a geographical north and south pole and another property of magnetic field is that magnetic field lines do not intersect.

34.(a) In the case of diamagnetic substances, the magnetic moments of atoms and the orbital magnetic moments have been oriented in such a manner that the vector sum of an atom's magnetic moment becomes zero. An external magnetic field can repel them weakly.

35.(d) Neil Bohr proposed a model, which is familiar as a planetary model of atoms. In Bohr's model, the neutrons and protons occupy a dense central region called the nucleus, and electrons orbit the nucleus much like planets orbiting the sun. Electrons are negatively charged and the nucleus is positively charged. Force in the former case is electrostatic force but in later case it is gravitational force.

36.(a) By using the right hand thumb rule, the direction of the magnetic field can be determined then by using Fleming's right hand rule the direction of force comes towards each other.

37.(d) As the galvanometer is used to check the current flow direction and the magnitude of the direct current. That's why the resistance of the galvanometer is nearly zero. This is somewhere similar to ammeter but both are different devices. Ammeter can only show us the current magnitude not the direction. A Galvanometer's needle can fluctuate in two directions whereas an ammeter's needle can only show one side deflection.

38.(b) Electric lines of forces do not exist inside a charged body, the magnetic lines of force travel from north to south pole and inside the magnet they are from south pole to north pole hence continuous.

39.(a) If the particle is moving along the direction of magnetic field, then $\theta = 0^\circ$ hence force becomes zero.

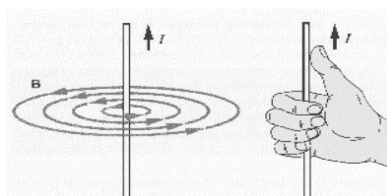
40. c) Assertion is true reason is false. Iron is a ferromagnetic material.

Case study based question

41.

1.(c) stationary charge creates electric field but moving charge creates both electric and magnetic field.

2.(a)



The direction of thumb is from south to north, using right hand thumb rule, if we curl our fingers on the left side, it shows the outward direction.

3.(d) Magnetic field is created around moving charge, a stationary charge cannot create magnetic field.

$$4.(b) F = ilB \sin \theta = 6 \times 10^{-2} \times 2 \times \sin 90^\circ = 120 \text{ N}$$

$$5.(a) \quad mg = ilB \sin \theta$$

$$100 \times 10^{-2} \times 9.8 = 5 \times 1 \times B \times \sin 90^\circ$$

$$B = 0.196 \text{ T}$$

SECTION B

MCQ

Q1. Two parallel conductors carrying 'i' current in the same direction what will be the force being experienced by each conductor (r = distance between the conductors)

(a) Repulsion and $\frac{\mu_0 2i}{4\pi r}$

(b) Repulsion and $\frac{\mu_0 i}{4\pi 2r}$

(c) Attraction and $\frac{\mu_0 i^2}{2\pi r}$

(d) Attraction and $\frac{\mu_0 i}{4\pi 2r}$

Q2. The magnetic field at a point midway between two parallel long wires carrying currents in the same direction is $10\mu T$. If the direction of the smallest current among them is reversed, the field becomes $30\mu T$. The ratio of the larger to the smaller current in them is

(a) 3 : 1

(b) 2 : 1

(c) 4 : 1

(d) 3 : 2

Q3. An electric current is flowing through a circular coil of radius R. The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is

(a) $2\sqrt{2}$

(b) 27

(c) 36

(d) 8

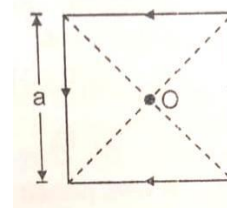
Q4. A square coil of side a carries a current l. The magnetic field at the centre of the coil is

(a) $\frac{\mu_0 l}{a\pi}$
 $\frac{2\sqrt{2}\mu_0 l}{a\pi}$

(b) $\frac{\sqrt{2}\mu_0 l}{a\pi}$

(c) $\frac{\mu_0 l}{\sqrt{2}a\pi}$

(d)



Q5. A current loop in a magnetic field

(a) Can be in equilibrium in one orientation

(b) Can be in equilibrium in two orientations, both the equilibrium states are unstable

(c) Can be in equilibrium in two orientations, one stable while the other is unstable

(d) Experiences a torque whether the field is uniform or non – uniform in all orientations

Q6. Force on a current carrying conductor in a magnetic field is

(a) $BIL \tan \Theta$

(b) $BI / L \sin \Theta$

(c) $BIL \sin \Theta$

(d) $BIL \cos \Theta$

Q7. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does not depend upon

- (a) Area of loop (b) Value of current (c) Magnetic field (d) None of these

Q8. What is the magnitude of magnetic force per unit length on a wire carrying a current of 4 amperes and making an angle of 30° with the direction of magnetic field of 2T?

- (a) 2 N/m (b) 3 N /m (c) 4 N/m (d) 5 N/m

Q9. We can convert moving coil galvanometer into voltmeter by:

- (a) Introducing resistance of large value in parallel.
(b) Introducing resistance of small value in series
(c) Introducing resistance of large value in series.
(d) Introducing resistance of small value in parallel

1 MARK QUESTIONS

Q10. In a certain region of space, electric field E and magnetic field B are perpendicular to each other. An electron enters in the region perpendicular to the directions of both B and E and moves undeflected. Find the velocity of the electron.

Q11. Two identical charged particles moving with same speed enter a region of uniform magnetic field. If one of these enters normal to the field direction and the other enters along a direction at 30° with the field. What would be the ratio of their angular frequencies?

Q12. An α particle and a proton are moving in the plane of the paper in a region where there is a uniform magnetic field $\vec{B}$ directed normal to the plane of the paper. If the two particles have equal linear momenta, what will be the ratio of the radii of their trajectories in the field?

Q13. A beam of protons projected along +x-axis experiences a force due to a magnetic field along -y-axis. What is the direction of the magnetic field?

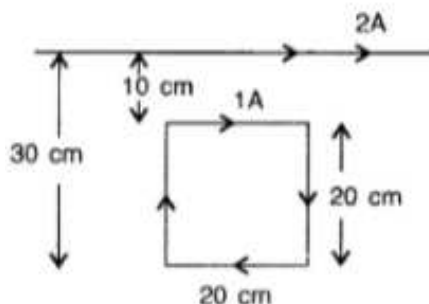
2 MARKS QUESTIONS

Q14. Can we decrease the range of an ammeter?

Q15. An electron and proton, moving parallel to each other in the same direction with equal momenta, enter into a uniform magnetic field which is at right angle to their velocities. Trace the trajectories in the magnetic field.

Q16. An ammeter of resistance $0.6\ \Omega$ can measure current up to 1.0 A . Calculate (i) The shunt resistance required to enable the ammeter to measure current up to 5.0 A (ii) The combined resistance of the ammeter and the shunt. (Delhi 2013)

Q17. A square loop of side 20 cm carrying current of 1 A is kept near an infinite long straight wire carrying current of 2 A in the same plane as shown in the figure. Calculate the magnitude and direction of the net force exerted on the loop due to the current carrying conductor.

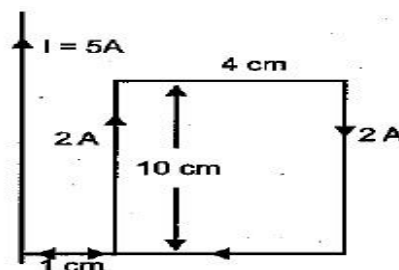


3-MARKS QUESTIONS

- Q18. (a) Write the expression for the magnetic force acting on a charged particle moving with velocity v in the presence of magnetic field B .
 (b) A neutron, an electron and an alpha particle moving with equal velocities, enter a uniform magnetic field going into the plane of the paper as shown. Trace their paths in the field and justify your answer.

5-MARKS QUESTIONS

Q19. A rectangular loop of wire as shown in the figure of size $4\text{ cm} \times 10\text{ cm}$ carries a steady current of 2 A . A straight long wire carrying 5 A current is kept near the loop as shown in fig. If the loop and the wire are coplanar find (a) Write an expression for torque. (b) The torque acting on the loop and (c) the magnitude and direction of the force on the loop due to the current carrying wire



Assertion-Reason

Select the correct answer to these questions from the codes (a), (b), (c) and (d) are as given below

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true but R is not the correct explanation of A.
 (c) A is true but R is false.
 (d) A is false and R is also false.

Q20. Assertion The poles of magnet cannot be separated by breaking into two pieces.

Reason The magnetic dipole moment will be reduced to half when broken into two equal pieces.

Q21. Assertion When a bar magnet is kept in an external uniform magnetic field, it starts oscillating.

Reason A restoring torque acts on the dipole when kept in the magnetic field.

Q22. Assertion Two parallel wires carrying currents in the opposite direction, attract each other.

Reason Parallel currents repel and antiparallel currents attract.

Q23. Assertion Gauss's theorem is not applicable in magnetism.

Reason Magnetic monopoles do not exist.

Q24. Assertion Magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason There is a uniform magnetic field inside the solenoid.

Q25. Assertion If a charged particle is projected in a region, where B is perpendicular to velocity of projection, then the net force acting on the particle is independent of its mass.

Reason The particle is performing rectilinear motion.

Q26. Assertion When a charged particle moves in a region of magnetic field such that its velocity is at some acute angle with the direction of field, its trajectory is a helix.

Reason Perpendicular component of velocity causes a rotating centripetal force and the parallel component of velocity does not produce any force.

Q27. Assertion For a current carrying wire loop of N turns, placed in a region of a uniform magnetic field B , the torque acting on it is given by $m \times B$.

Reason Whenever the magnetic moment m is perpendicular to B , then torque on the loop will be zero.

Q28. Assertion The current sensitivity of a galvanometer is the deflection of current per unit current passing through the coil.

Reason The galvanometer can be used as a detector to check if a current is flowing in the circuit.

Q29. Assertion Magnetic field lines always form closed loops.

Reason Moving charges or currents produce a magnetic field.

Case study based question

Q30. Magnetic moment

The directional property of magnets was also known since ancient times. A thin long piece of a magnet, when suspended freely, pointed in the north-south direction. Magnetic field is responsible for the most notable property of a magnet. It is a force that pulls on other ferromagnetic materials, such as iron, steel, nickel, cobalt, etc., and attracts or repels other magnets. Magnet's magnetic moment is a vector that characterises the magnet's overall magnetic properties. It is also called magnetic dipole moment and usually denoted by **m**.

For a bar magnet, the direction of the magnetic moment points from the magnet's south to north-pole and the magnitude relates to how strong and how far apart these poles are.

1. In a uniform magnetic field the net magnetic force on the dipole

- (a) Is always zero
- (b) Depends on the orientation of the dipole
- (c) Can never be zero
- (d) Depends on the strength of the dipole

2. Torque acting on a magnetic dipole in uniform magnetic field at an acute angle is

- (a) Zero
- (b) Nonzero
- (c) Equal to force
- (d) None of these

3. The magnetic dipole moment of a circular coil current carrying I and area A is

- (a) IA
- (b) NIA
- (c) $\mu_0 nI$
- (d) $nIAB$

4. Which of the following cannot behave as a magnetic dipole

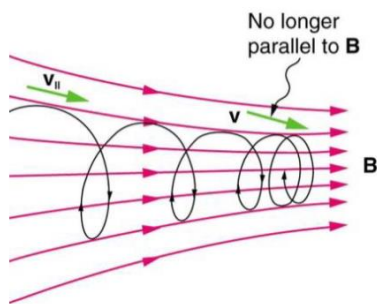
- (a) Electron revolving around nucleus
- (b) Current loop
- (c) Diamagnetic materials
- (d) All of them

5. The ultimate individual unit in any magnet is a

- (a) south-pole
- (b) north-pole
- (c) Quadrupole
- (d) Dipole

Q 31. FORCE ON A CHARGE IN ELECTRIC AND MAGNETIC FIELD

A point charge q (moving with a velocity v and located at r at a given time (t) in the presence of both the electric field E and magnetic field B . The force on an electric charge q due to both of them can be written as $F = q [E + v \times B] = F_{el} + F_{mag}$. It is called the 'Lorentz force'.



1. If the charge q is moving under a field, the force acting on the charge depends on the magnitude of field as well as the velocity of the charge particle, what kind of field is the charge moving in?

- (a) Electric field (b) Magnetic field
(c) Both electric and magnetic field perpendicular to each other (d) None of these

2. The magnetic force acting on the charge ' q ' placed in a magnetic field will vanish if

- (a) if v is small (b) If v is perpendicular to B (c) If v is parallel to B
(d) None of these

3. If an electron of charge $-e$ is moving along $+X$ direction and magnetic field is along $+Z$ direction, then the magnetic force acting on the electron will be along

- (a) $+X$ axis (b) $-X$ axis (c) $-Y$ axis (d) $+Y$ axis

4. The vectors which are perpendicular to each other in the relation for magnetic force acting on a charge particle are

- (a) F and v (b) F and B (c) v and B (d) All of these

5. A particle moves in a region having a uniform magnetic field and a parallel, uniform electric field. At some instant, the velocity of the particle is perpendicular to the field direction. The path of the particle will be

- (a) A straight line (b) A circle (c) A helix with uniform pitch (d) A helix with non-uniform pitch

ANSWER KEY

SECTION B

MCQ

1. (c) Attraction and $\frac{\mu_0 i^2}{2\pi r}$
2. (b) 2 : 1
3. (b) 27
4. (d) $\frac{2\sqrt{2}\mu_0 l}{a\pi}$
5. (c) Can be in equilibrium in two orientations, one stable while the other is unstable
6. (c) $BIL\sin\theta$
7. (d) None of these
8. (c) 4N/m hint: $BIL\sin\theta$
9. (c) introducing resistance of large value in series

1 MARK

10. Net force on electron moving in the combined electric field E and magnetic field B is

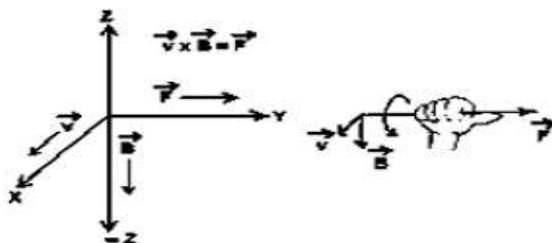
$$F = e(E + V \times B)$$

Since electron moves undeflected then $F = 0$

$$e(E + V \times B) = 0$$

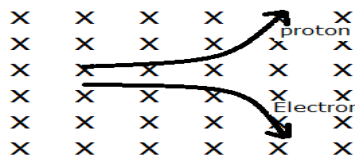
$$V = E / |B|$$

11. 1:1, The angular frequency does not depend on the speed of the particle or radius of its orbit.
12. 1:2 ($r = mv/qB = p/qB$, for equal p , $r \propto 1/q$)
13. The direction of Magnetic field is towards positive direction of z axis.



2-MARKS

14. An ammeter is basically a permanent magnet moving coil (PMMC) instrument which deflects for very small amount of current (in mA range). Its only possible to increase the range of an ammeter but we cannot decrease the range of an ammeter. The range of an ammeter can be easily increased by adding a shunt resistance of very low value to bypass the major part of the current through the resistance path instead of ammeter. This increases the range of an ammeter by reducing the actual current flow through the ammeter. There is a way to decrease the range of an ammeter but it is not practically feasible. If we decrease the restoring torque by reducing spring stiffness inside the ammeter it will decrease the range of the ammeter. We are basically increasing the sensitivity of the ammeter. But reducing spring stiffness causes large stray errors in the readings and is therefore not practically feasible.
15. A magnetic field does not exert any force on a charge moving parallel or antiparallel to the field direction. Since they are travelling in the direction of the magnetic field, there will be no force acting on them. Hence their paths will remain the same after entering the magnetic field.



16. (i) Shunt resistance, $S = I_g G / I - I_g$

$$1 \times 0.6 / (5-1) \\ 0.15 \text{ ohm}$$

- (ii) Total resistance,

$$1/R = 1/6 + 1/15 \\ = (50 + 200) / 30 \\ = 250/30$$

$$\text{Therefore, } R = 30/250 = 0.12 \text{ Ohms}$$

17. We are given:

$$I_1 = 2\text{A}, I_2 = 1\text{A}$$

$$r_1 = 10 \text{ cm}, r_2 = 30 \text{ cm}$$

$$\mu_0 = 4 \pi \times 10^{-7} \text{ TmA}^{-1}$$

We have

Now net force on the side will be;

$$F = \mu_0 I_1 I_2 l (1/r_1 - 1/r_2)$$

$$F = 2 \times 10^{-7} \times 1 \times 2 \times (20 \times 10^{-2}) [1/10 \times 10^{-2} - 1/30 \times 10^{-2}]$$

$$F = 5.33 \times 10^{-7} \text{ N}$$

The direction of force is towards the infinitely long straight wire.

3-MARKS

18. Lorentz force = magnetic force + electric force

$$F = [Qvb \sin\theta + Qe]$$

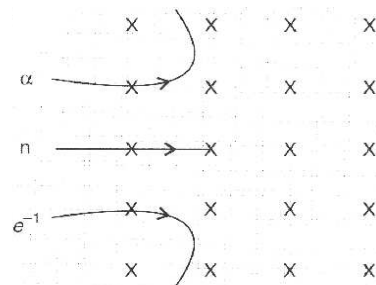
We know that a charged particle will experience a force when it enters a magnetic field. The magnetic field will move the charged particle in a circular path, as the force is perpendicular to the velocity of particle. The radius of the circular path will be given by

$$mv^2 / r = Bqv$$

$$r = mv / Bq$$

As B and v are constant, we can write

$$r \propto m/q$$



The neutron will move along the straight line as it has no charge.

The electron will inscribe a circle of radius smaller than that of the alpha particle as the mass to charge ratio of the alpha particle is more than that of the electron. So, the alpha particle will move in the clockwise direction and the electron will move in anticlockwise direction according to the right-hand rule.

5 MARKS

19. (a) $\tau = MB \sin\theta = \tau = M \times B$

(b) $\tau = M \times B = Mbsin\theta$

here M and B are in the same direction so

$$\theta = 0, \therefore \tau = 0$$

© we know, $F_B = \odot l \times B$

Force between two current carrying wires is given by

$$F = (\mu_0/2\pi) \cdot (I_1 I_2/r) \times l = (\mu_0/2\pi) \times (5 \times 2/1 \times 10^{-2}) \times 0.10$$

$$F = (\mu_0/2\pi) \cdot 100$$

$$\text{And } F' = (\mu_0/2\pi) \times (5 \times 2/1 \times 10^{-2}) \times 0.10$$

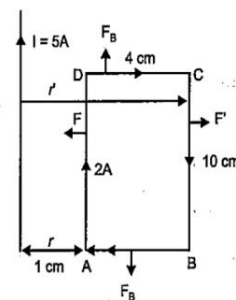
$$F' = (\mu_0/2\pi) \times 20$$

Now, resultant force on the loop,

$$F_{\text{net}} = F - F' = (\mu_0/2\pi) (100 - 20)$$

$$= 2 \times 10^{-7} \times 80 = 16 \times 10^{-6} \text{ N}$$

The direction of net force is towards the straight wire i.e., attractive.



Assertion-Reason

20.(b) The magnetic dipole moment will be reduced to half when broken into two equal pieces and every atom behaves like a dipole so the dipole of a magnet cannot be separated.

21.(a) When a bar magnet is placed in a uniform magnetic field. Then the bar magnet will experience only torque and no force, and this torque on the bar magnet will be acting on both ends, and will be equal but opposite in direction.

22.(d) Parallel currents attract and antiparallel currents repel.

23.(a) Gauss's law of magnetism is different from that for electrostatics because electric charges do not necessarily exist in pairs but magnetic monopoles do not exist.

24.(b) $B = \mu_0 Ni$, from this formula we see the dependence of B in current and inside a solenoid it is uniform.

25.(c) Force $= q(\mathbf{v} \times \mathbf{B})$ it is independent of mass and if $\mathbf{v}$ and $\mathbf{B}$ are perpendicular to each other, the particle describes a circle.

26.(a) A charged particle moves in a circle when its velocity is perpendicular to the magnetic field. When it forms an acute angle with the magnetic field, it can be resolved in two components, parallel and perpendicular. The perpendicular components tend to move it in a circle, the parallel components tend to move along the magnetic field to form a helical motion of uniform radius and pitch.

27.(c) Torque, $\tau = mB\sin\theta$

Here $\theta = 90^\circ$, $\sin 90^\circ = 1$ so torque will be maximum.

28.(b) A galvanometer is a device that is used to detect small electric current or measure its magnitude. The current sensitivity of a galvanometer is the deflection of current per unit current passing through the coil $I_s = \frac{NAB}{k}$

29.(b) Assertion is the property of a magnet while reason is one of the sources of magnetic field.

Case study based question

30. 1.(a) The forces of magnitude Mb act opposite to each other on two poles, hence net force acting on the dipole due to external magnetic field is zero.

2.(b) Forces acting on the poles are along different lines of action constituting a couple hence non zero torque.

3.(a) The magnetic dipole moment of a current loop is the product of the current passing through the loop and the area inside the loop.

4.(c) Materials with no unpaired, or isolated electrons are considered diamagnetic. Diamagnetic substances do not have magnetic dipole moments.

5.(d) Every magnet has two poles, no monopole exists.

31.1. (b) Magnetic field

2.(c) If v is parallel to B

3.(d) + Y axis

4.(d) All of these

5.(d) A helix with non-uniform pitch

SECTION C

MCQ

Q1. The magnetic force acting on a charged particle of charge $-2\mu C$ in a magnetic field of $2T$ acting in y direction, when the particle velocity is $(2\hat{i} + \hat{j}) \times 10^6 ms^{-1}$, is

- (a) 4 N in z direction
(b) 8 N in y direction
(c) 8 N in z direction
(d) 8 N in $-z$ direction

Q2. Two long parallel wires carry currents i_1 and i_2 such that $i_1 > i_2$. When the currents are in the same direction, the magnetic field at a point midway between the wires is $6 \times 10^{-6}T$. If the direction of i_2 is reversed, the field become $3 \times 10^{-5}T$. The ratio of $\frac{i_1}{i_2}$ is

- (a) $1/2$ (b) 2 (c) $2/3$ (d) $3/2$

Q3. A straight wire carrying current I is made in circular loop. If M is the magnetic moment associated with the loop, then the length of the wire is

- (a) $\sqrt{\frac{4\pi M}{I}}$ (b) $\sqrt{\frac{2\pi M}{I}}$ (c) $\sqrt{\frac{\pi M}{I}}$ (d) $\sqrt{\frac{\pi M}{2I}}$

Q4. Two similar coils of radius R are lying concentrically with their planes at right angles each other. The currents flowing in them are I and $2I$, respectively. The resultant magnetic field induction at the centre will be

- (a) $\frac{\sqrt{5}\mu_0 I}{2R}$ (b) $\frac{3\mu_0 I}{2R}$ (c) $\frac{\mu_0 I}{2R}$ (d) $\frac{\mu_0 I}{R}$

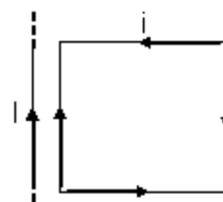
Q5. An electron is projected with uniform velocity along the axis of a current carrying long solenoid. Which of the following is true?

- (a) The electron will be accelerated along the axis.
(b) The electron path will be circular about the axis.
(c) The electron will experience a force at 45° to the axis and hence execute a helical-path.
(d) The electron will continue to move with uniform velocity along the axis of the solenoid.

Q6. In a certain region of space electric field $\vec{E}$ and magnetic field $\vec{B}$ are perpendicular to each other. An electron enters perpendicularly to both fields and moves undeflected. The velocity of electron is

- (a) E/B (b) B/E (c) $\vec{E} \times \vec{B}$ (d) $\vec{E} \cdot \vec{B}$

Q7. A rectangle loop carrying a current (i) is situated near a long straight wire such that the wire is parallel to the one of the sides of the loop and is in the plane of the loop. If a steady current i is established in wire as shown in the figure below, the loop will



- (a) Rotate about an axis parallel to the wire (b) Move away from the wire or towards right
(c) Move towards the wire or towards left (d) Remain stationary

Q8. A wire in the form of a circular loop, of one turn carrying a current, produces magnetic field B at the centre. If the same wire is looped into a coil of two turns and carries the same current, the new value of magnetic induction at the Centre is

- (a) B (b) $2B$ (c) $4B$ (d) $8B$

Q9. When a magnetic dipole of moment $\vec{M}$ rotates freely about its axis from unstable equilibrium to stable equilibrium in a magnetic field $\vec{B}$, the rotational kinetic energy gained by it is :

- (a) $-M.B$ (b) $\frac{2}{M.B}$ (c) $2M.B$ (d) $\frac{M}{B}$

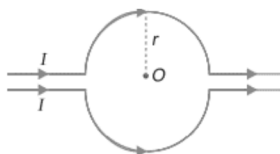
Q10. The current sensitivity of a galvanometer increases by 20%. If its resistance also increases by 20%, the voltage sensitivity will be

- (a) Decrease by 1% (b) Increased by 10% (c) Increased by 5% (d) Decrease by 4%

1 MARK QUESTIONS

Q11. An electron, passing through a region is not deflected. Are you sure that there is no magnetic field in that region?

Q12. What is the value of magnetic field at point O due to current flowing in the wires?



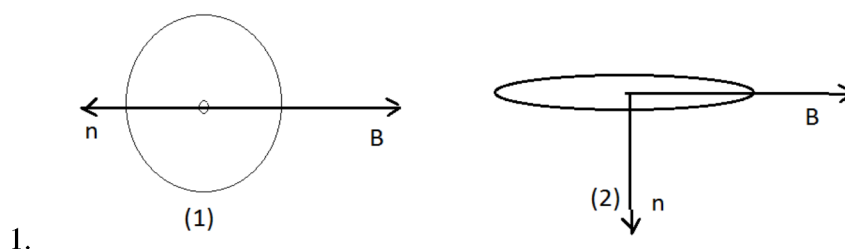
Q13. A long straight wire carries a steady current I along the Positive y -axis in a co-ordinate system. A particle of charge $+Q$ is moving with a velocity $\vec{v}$ along the x -axis. In which direction will the particle experience a force?

2-MARK QUESTIONS

Q14. A wire AB is carrying a steady current of 6A and is lying on the table. Another wire CD carrying 4A is held directly above AB at a height of 1mm. Find the mass per unit length of the wire CD so that it remains suspended at its position when left free. Give the direction of the current flowing in CD with respect to that in AB. [Take the value of $g=10\text{ms}^{-2}$] [All India 2013]

Q15. A galvanometer of resistance 'G' can be converted into a voltmeter of range (0-V) volts by connecting a resistance 'R' in series with it. How much resistance will be required to change its range from 0 to $v/2$?

Q16. A current loop is placed in a uniform magnetic field in the following orientations (1) and (2). Calculate the magnetic moment in each case.



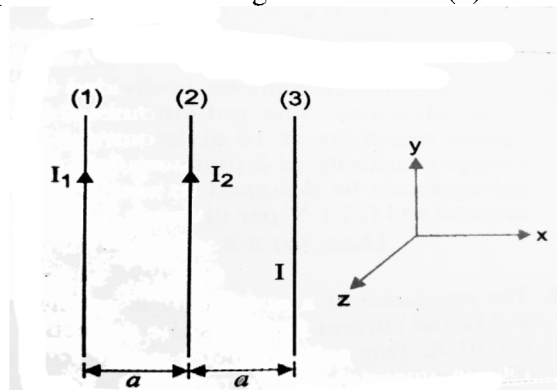
3-MARKS QUESTIONS

Q17. Three long straight parallel wires are kept as shown in the figure. The wire (3) carries a current I

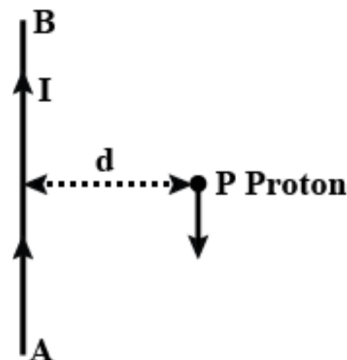
(a) The direction of the flow of current I in the wire 3, is such that the net force on wire 1 due to the other two wires is zero

(b) By reversing the direction of current I, the net force on wire 2 due to the other two wires becomes zero. What will be the direction of

the current I in the two cases? Also obtain the magnitudes of currents I_1 , I_2 and I.



Q18. A long straight wire AB carries a current I . A proton P moves with a speed V , parallel to the wire at a distance d from it in a direction opposite to the current. What is the force experienced by the proton and what is its direction?



5-MARK QUESTIONS

- Q19. (a) State the principle of working of a galvanometer
 (b) A galvanometer of resistance G is converted into a voltmeter to measure upto V volts by connecting a resistance R_1 in series with the coil. If a resistance R_2 is connected in series with it then it can measure upto $V/2$ volts. Find the resistance, in terms of R_1 and R_2 , required to be connected to convert it into a voltmeter that can read upto $2V$. Also find the resistance G of the galvanometer in terms of R_1 and R_2 .
 (c) "Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity." Justify the statement.

Assertion-Reason

Select the correct answer to these questions from the codes (a), (b), (c) and (d) are as given below

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true but R is not the correct explanation of A.
 (c) A is true but R is false.
 (d) A is false and R is also false.
20. **Assertion:** Steady current is the only source of magnetic field.
Reason: Only moving charge can create magnetic field.
21. **Assertion:** A magnetic field does not interact with a stationary charge.
Reason: A moving charge produces a magnetic field.
22. **Assertion:** When velocity of electron is perpendicular to B it will perform circular motion.
Reason: Magnetic force is perpendicular to velocity.
23. **Assertion:** A beam of electrons can pass undeflected through a region of E and B .
Reason: Force on moving charged particles due to magnetic field may be zero in some cases.

24. **Assertion:** If the path of a charged particle in a region of uniform electric and magnetic field is not a circle, then its kinetic energy will not remain constant.

Reason: In a combined electric and magnetic field region, a moving charge experiences a net force $F = qE + q(v \times B)$, where symbols have their usual meanings.

25. **Assertion:** If we increase the current sensitivity of a galvanometer by increasing the number of turns, its voltage sensitivity also increases.

Reason: Resistance of a wire also increases with N .

26. **Assertion:** When a magnetic dipole is placed in a non uniform magnetic field, only a torque acts on the dipole.

Reason: Force would not act on dipole if magnetic field were non uniform.

27. **Assertion:** Galvanometer can as such be used as an ammeter to measure the value of the current in given circuit

Reason: It gives a full-scale deflection for a current of the order of ampere.

28. **Assertion:** Diamagnetic materials can exhibit magnetism.

Reason Diamagnetic materials have permanent magnetic dipole moments.

29. **Assertion:** Paramagnetic materials can exhibit magnetism.

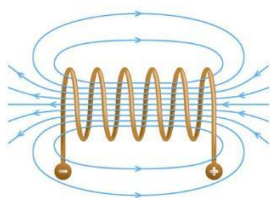
Reason: Paramagnetic materials have permanent magnetic dipole moments.

Case study based question

Q30.

Magnetic Field due to a Solenoid

Seema wound a very long insulated copper wire on a plastic pipe and carefully took the pipe out. The two ends of the copper wires are then attached to a battery. This cylindrical shape of copper wire is called a solenoid and this solenoid is used in several devices such as door bell, door locks, speakers etc. The magnetic field of a solenoid is given below



The magnetic field strength of a solenoid having n turns is $\mathbf{B} = \mu_0 n I$, where, I is the current flowing in the solenoid, n is number of turns per unit length and μ_0 is the permeability of free space.

1. A long solenoid has 400 turns per meter and it is used as an electromagnet. If 1.5A current is flowing through it, what is the strength of the electromagnet

- (a) 0.008 T (b) 0.4 T (c) 0.6 T (d) 0.007 T

2. Solenoid has length l , N turns and carrying a current I , what will be the magnetic field inside the solenoid

- (a) $\mu_0 n I$ (b) $\frac{\mu_0 N I}{l}$ (c) $\mu_0 n l^2$ (d) None of these

3. The strength of magnetic field in a solenoid cannot be affected by

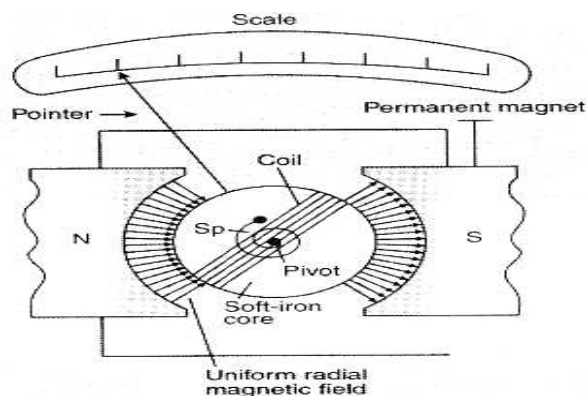
- (a) Increasing its length (b) Decreasing the value of current
(c) Decreasing the number of turns (d) None of these

4. The strength of magnetic field outside a solenoid is

- (a) Infinity (b) Zero
(c) Double the value of field inside
(d) Half the value of the field inside

5. The nature of magnetic field lines passing through the current carrying solenoid is

- (a) Closed loop
(b) Discontinuous curve
(c) Straight line
(d) None of these



SOURCE BASED QUESTION

Q31. Analog voltmeters and ammeters work by measuring the torque exerted by a magnetic field on a current carrying coil. The reading is displayed by means of the deflection of a pointer

over a scale. The adjacent figure shows the essentials of a galvanometer, on which both Analog ammeters and Analog voltmeters are based. Assume that the coil is 2.1 cm high, 1.2 cm wide has 250 turns and is mounted so that it can rotate about an axis (into the page) in a uniform radial magnetic field with $B = 0.23 \text{ T}$. for any orientation of the coil, the net magnetic field through the coil is perpendicular to the normal vector of the coil (and thus parallel to the plane of coil) A spring SPSP provides a counter torque that balance the magnetic torque so that a given steady current I in the coil results in a steady angular deflection ϕ . The greater the current is greater the deflection is and thus greater the torque required of the spring is A current of $100 \mu\text{A}$ produces an angular deflection of 28°

1. What must be the torsional constant K of the spring?

- (a) $2.6 \times 10^{-8} \text{ Nm/degree}$ (b) $5.2 \times 10^{-8} \text{ Nm/degree}$
 (c) $2.6 \times 10^{-4} \text{ Nm/degree}$ (d) $5.2 \times 10^{-6} \text{ Nm/degree}$

2. If we reduce the value of this K to half of its value then the deflection would be

- (a) 28° (b) 56° (c) 14° (d) none

3. If the value of magnetic field is put equal to 0.69 T and $K = 15.6 \times 10^{-8} \text{ Nm/degree}$. Then the deflection would be

- (a) $< 28^\circ$ (b) $= 28^\circ$ (c) 14° (d) none

ANSWER KEY

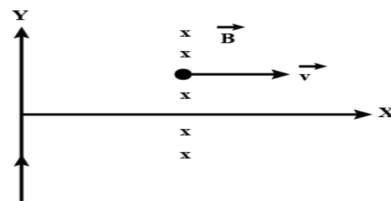
SECTION C

MCQ

1. (d) 8 N in $-z$ direction
2. (d) $3/2$
3. (a) $\sqrt{\frac{4\pi M}{l}}$
4. (a) $\frac{\sqrt{5}\mu_0 l}{2R}$
5. (d) The electron will continue to move with uniform velocity along the axis of the solenoid.
6. (a) E/B (hint: $qE = qvB$)
7. (c) Move towards the wire or towards left
8. (c) $4B$
9. (c) $2MB$
10. (d) decrease by 4%

1-MARK

11. No, if an electron enters parallel to a magnetic field, no force acts and the electron remains undeflected.
12. Zero, because the upper and lower current carrying conductors are identical and so the magnetic fields caused by them at the centre O will be equal and opposite.
13. From Fleming's left-hand rule, thumb points along + y direction, so the direction of magnetic force will be along + y axis (or in the direction of flow of current).



14. Weight per unit length of the wire must be supported by magnetic force per unit length.

$$(m/l) g = \mu_0 I_1 I_2 / 2\pi r$$

$$(m/l) = (\mu_0 I_1 I_2 / 2\pi r) g$$

$$= 4\pi \times 10^{-7} \times 6 \times 4 / 2\pi (10^{-3}) (10)$$

$$= 2 \times 10^{-7} \times 6 \times 4 / 10^{-3} \times 10$$

$$= 48 \times 10^{-5} \text{ kg/m}$$

$$= 0.48 \text{ mg/m.}$$

15. Given: resistance of galvanometer = $G \Omega$

Range of voltmeter (RL) = $(0-V)$ volts

Resistance to be connected in parallel = R

$R'=?$ where range is $(0-V/2)$ volts

In the first case $i_g = V / (R+G)$ (i)

In the second case $i_g = V/2 / (R'+G)$ (ii)

[i_g is the maximum current which can flow through galvanometer]

From equation (i) and (ii) on solving we get

$$R' = (R-G) / 2$$

16. (a) -mB (b) Zero

3 MARKS

17.(a) Net force experienced by wire (1) can be zero only when the current in wire (3) flows along -J i.e. downwards it means that force acting on the wire (1) due to wire (3) and wire (2) are equal and opposite

$$\mu_0 I_1 I / 2\pi(2a) = \mu_0 I_1 I_2 / 2\pi(a)$$

$$\text{Therefore, } I = 2I_2$$

(b) when direction of current in wire (3) is reversed then current should be along +j i.e., upwards. For this case net force on wire (2) becomes zero, which means that the force due to wire (1) and wire (3) are equal and opposite

$$\mu_0 I_1 I_2 / 2\pi a = \mu_0 I_2 I / 2\pi(a)$$

$$\text{therefore } I = I_1$$

$$I = I_1 = 2 I_2$$

18. Magnetic field due to the straight wire AB at a perpendicular, distance r from it.

$$B = \mu_0 I / (2\pi r)$$

Therefore, force on proton moving with velocity 'v' perpendicular to B, is

$$F = qvB = \mu_0 I q v / (2\pi r)$$

The direction of force on proton acts in plane of paper towards right.

5 MARKS

19. (a) Galvanometer works on the principle of torque on a coil carrying current in a magnetic field. The torque acting on the coil is proportional to the magnitude of electric current. The coil is connected to a pointer and the pointer of the galvanometer is calibrated to show correct deflection on a scale.

(b) Let the resistance of the galvanometer be R_G .

Let the max. current carrying capacity be I .

From the given data,

$$V = I (R_G + R_1)$$

$$V/2 = I (R_G + R_2)$$

From the equations,

$$2 = (R_G + R_1) / (R_G + R_2)$$

$$R_G = R_1 - R_2$$

Let the resistance required to read $2V$ be R_3

$$2V = I (R_G + R_3)$$

$$2 (R_G + R_1) = R_G + R_3$$

$$R_3 = R_G + 2R_1$$

$$R_3 = 3R_1 - 2R_2$$

(c) Due to deflecting torque, the coil rotates and suspension wire gets twisted. A restoring torque is set in the suspension wire.

$$NIBA = k\theta \quad \text{(In equilibrium)}$$

$$I = k / (NBA) \cdot \theta$$

$$I = G\theta$$

Where, $G = k / (NBA)$

It is known as galvanometer constant.

$$\theta/I = NBA / k$$

Therefore, current sensitivity of the galvanometer is the deflection per unit current.

$$\theta/V = (NBA/k). (I/V) = (NBA/k). 1/R$$

Therefore, voltage sensitivity is the deflection per unit voltage. It depends on the resistance also.

Assertion-Reason

20.(d) moving charge, electron, current carrying wire, naturally occurring magnet all these can create magnetic field.

21.(a) Stationary charge doesn't get affected by magnetic field, moving charge creates magnetic field.

22.(a) If $F \perp V$, at all instants then motion will be circular.

23.(b) For velocity selector the electrons can pass undeflected, when velocity and magnetic field are parallel, in that case force can be zero, both are correct but reason is not correct explanation of assertion.

24.(a) $F = qE + q(v \times B)$ this is called Lorentz force.

Due to electric field, acceleration $a = \frac{qE}{m}$, hence velocity will not remain constant.

25.(c) current sensitivity $I_s = \frac{\phi}{I} = \frac{NAB}{k}$

voltage sensitivity $V_s = \frac{NAB}{kR}$

$$V_s = \frac{I_s}{R}$$

26. (d) In a non-uniform magnetic field both the torque and force will act.

27.(d) Galvanometer is used to detect the direction of current in a circuit, it gives full scale deflection to the current of microampere range.

28.(c) Diamagnetic materials get repelled in magnetic fields so they exhibit magnetism but they don't have unpaired electrons so we can say that they don't have dipole moment.

29.(a) Paramagnetic materials have unpaired electrons so they exhibit magnetism.

Case study based

30. 1.(d) $B = \mu_0 nI$

where $n = 400$ and $I = 1.5A$

putting values $B = 0.007 T$

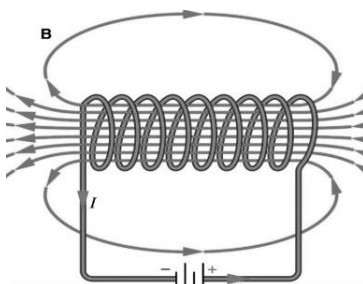
2. (b) Derivation of magnetic field inside a solenoid.

3. (d) $\mathbf{B} = \mu_0 n \mathbf{I}$

where n is number of turns per unit length.

4. (b) The magnetic field outside the solenoid is so weak that it is taken to be zero.

5. (a)



The magnetic field lines in a solenoid form closed loop.

SOURCE BASED QUESTION

31. Due to deflecting torque, the coil rotates and suspension wire gets twisted. A restoring torque is set in the suspension wire.

$$\tau = MB \sin 90^\circ$$

$$= NIAB$$

$$\text{Also, } \tau = K \theta$$

$$K = \tau / \theta = NIAB / \theta = (250 \times 100 \times 10^{-6} \times 2.1 \times 1.2 \times 10^{-4}) / 28$$

$$= 5.2 \times 10^{-8} \text{ Nm/degree}$$

$$(b) 5.2 \times 10^{-8} \text{ Nm/degree}$$

$$2. (b) 56^\circ \text{ using } \tau = K \theta$$

$$3. (b) = 28^\circ \text{ using } \theta = \tau / K$$
